INCH-POUND

MIL-C-52032D(AV)
6 April 1995
SUPERSEDING
MIL-C-52032C
22 September 1975

MILITARY SPECIFICATION

COATING, INTERIOR, FUEL RESISTANT, VOC COMPLIANT FOR GASOLINE CONTAINERS

This specification is approved for use within the US Army Aviation and Troop Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a thermal cure coating with a maximum Volatile Organic Compound (VOC) content of 420 grams per liter (3.5 pounds per gallon) to be used on the interior of steel fuel containers.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY AVIATION & TROOP CMD, ATTN AMSAT R EDS, ST LOUIS MO 63120-1798, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8010

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:05Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings.
PPP-P-1892	- Paint, Varnish, Lacquer and Related Materials Packaging, Packing and Marking of.

MILITARY

MIL-T-83133	- Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8) and NATO F-35.

STANDARDS

FEDERAL

FED-STD-141	- Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing.
FED-STD-313	- Material Safety Data Sheets Preparation and the Submission of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: STDZN DCMNT, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1729	- Visual Evaluation of Color Differences of Opaque Materials.
D 1849	- Standard Test Method for Package Stability of Paint.
D 2794	- Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
D 3335	- Standard Test Method for Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy.
D 3960	- Determining Volatile Organic Content (VOC) of Paints and Related Coatings, Standard Practice For.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T21:05Z
Check the source to verify that this is the current version before use.

E 1347 - Standard Test Method for Color and Color Difference Measusrement by Tristimulus (Filtered) Colorimetry.

(Application for copies should be addressed to the AMERCN SCTY & MTRLS 1916 RACE STRET, PHILADELPHIA PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The manufacturer is given wide latitude in the selection of the component materials to produce a coating formulation which meets the requirements of this specification.

3.1.1 Toxic products and formulations. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service which will act as an advisor to the contracting agency.

3.1.2 Prohibited material. The product shall not contain benzene, chlorinated solvents or ethylene based glycol ethers and their acetates. The lead content shall not exceed 0.06 percent of the non volatile content. Chromium (hexavalent) compounds shall not be used.

3.2 Condition in container. When examined as specified in 4.3.3.3, the coating shall show no evidence of putrefaction, hard settled pigment, or corrosion of the container. The coating shall be dispersible to a uniform condition by 5 minutes hand stirring without decanting or remixing.

3.3 Accelerated storage stability. After storage as specified in 4.3.3.4, a sealed, full one quart can of coating shall show no coagulation or hard settled pigment. The coating shall be dispersible to a uniform condition by 5 minutes hand stirring and shall pass the spraying properties tests specified in 3.4.

3.4 Spraying properties. When tested as specified in 4.3.2.2, the coating shall show no running, sagging, streaking or orange peel. The dried film shall show no dusting, mottling, color separation, flooding or floating.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:05Z
Check the source to verify that this is the current version before use.

3.5 Impact resistance. When tested as specified in 4.3.2.3, the coating shall sho no cracking or loss of adhesion.

3.6 Flexibility. When tested as specified in 4.3.2.4, the coating shall show no cracking, either at room temperature or at low temperature.

3.7 Resistance to boiling water. When tested as specified in 4.3.2.5, the coating shall show no blistering, softening, lifting, loss of adhesion, cracking, whitening, evidence of rusting of the panel or other differences in appearance from the coating on the non-immersed panel.

3.8 Inertness. When tested as specified in 4.3.2.6, the total gain or loss in weight of any coupon shall not exceed 3.0 milligrams.

3.9 Color. The cured coating shall be any light color. When tested as specified in 4.3.2.7, the Y value of directional reflectance shall be not less than 30 percent.

3.10 Opacity. When tested as specified in 4.3.2.8, the coating shall be opaque.

3.11 Volatile organic compounds (VOC). When tested as in 4.3.3.5 the VOC limit shall not exceed 420 gms/liter (3.5 lbs/gallon).

3.12 Fuel and water resistance. When tested as in 4.3.2.9, the panels, upon removal, shall show no softening or loss of adhesion. A moderate darkening of the area immersed in JP-8 is acceptable. There shall be no blistering to a greater extent than three random small blisters. Blister formation within 1/4 inch of the panel edge shall be ignored.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order and unless disapproved by the Government, the contractor's own facilities or any other, suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:05Z
Check the source to verify that this is the current version before use.

the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Material Safety Data Sheet. Material Safety Data Sheets not prepared in accordance with FED-STD-313 shall be cause for rejection. The contractor shall overpack a copy of the MSDS with each shipment of material.

4.2 Sampling and inspection. Unless otherwise specified, sampling, inspection, and testing shall be in accordance with FED-STD-141, method 1000.

4.2.3 Inspection of preparation for delivery. Packaging, packing, marking and palletization (as applicable) shall be inspected in accordance with the quality assurance provisions in PPP-P-1892.

4.3 Test methods.

4.3.1 Test conditions. The testing conditions shall be in accordance with FED-STD-141, section 9 or in accordance with the appropriate ASTM method except as otherwise specified herein. All test specimens shall be prepared and tested at a temperature of 20 to 30 °C (68 to 86 °F) and a relative humidity of 40 to 65 percent. Failure of any test result to fall within the ranges specified in section 3 as applicable, shall constitute failure of the applicable test.

4.3.1.1 Tests procedures. The following tests (see table I) shall be conducted in accordance with FED-STD-141, ASTM, or as specified herein. The right is reserved to make any additional tests deemed necessary to determine that the coating meets the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:05Z
Check the source to verify that this is the current version before use.

TABLE I. Tests.

Test	Reqmnt Para	Test Para	ASTM Method	FED-STD-141 Method
Chromium (Hexavalent) Content	3.1.2	4.3.3.1	--	--
Lead Content	3.1.2	4.3.3.2	D 3335	--
Condition in Container	3.2	4.3.3.3	--	3011
Accelerated Storage Stability	3.3	4.3.3.4	D 1849	--
Spraying Properties	3.4	4.3.2.2	--	4331
Impact Resistance	3.5	4.3.2.3	D 2794	--
Flexibility	3.6	4.3.2.4	--	--
Resistance to boiling water	3.7	4.3.2.5	--	--
Inertness	3.8	4.3.2.6	--	--
Reflectance	3.9	4.3.2.7	E 1347	--
Opacity	3.10	4.3.2.8	--	--
VOC determination	3.11	4.3.3.5	D 3960	--
Fuel and water resistance	3.12	4.3.2.9	--	--

4.3.2 Tests using laboratory prepared panels. Metal panels used for test purposes shall be of three types:

a. cold rolled steel, 3 by 6 inches, 0.032 inches thick.[1]
b. cold rolled steel, 3 by 6 inches. 0.010 inches thick.[2]
c. steel, tinplated, 3 by 6 inches, 0.010 inches thick.[3]

4.3.2.1 Preparation of panels. Panels shall be surface cleaned and phosphate coated in accordance with TT-C-490, type I. Then the coating under test shall be applied, dried and cured in accordance with the coating manufacturer's directions to give a dry film thickness of 0.7 ±0.1 mil, except the panel for the opacity test. All coated panels shall be free of fatty edges, sags or other application defects. Coated and cured panels for all tests shall be aged for 24 hours at 68 to 86 °F and 40-65 percent relative humidity, before testing.

4.3.2.2 Spraying properties. Holding the spray gun 8-10 inches from the panel, spray to a dry film thickness of 0.7 ±0.1 mil according to method 4331 of FED-STD-141. Cure coating according to manufacturer's instructions and evaluate for compliance to 3.4.

---

[1]/ Q-Panel Company, Cleveland, Ohio,R-36, or equivalent.

[2]/ Q-Panel Company, Cleveland, Ohio, D-36, or equivalent.

[3]/ Q-Panel Company, Cleveland, Ohio, tinplated, DT 36, or equivalent.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:05Z
Check the source to verify that this is the currentversion before use.

4.3.2.3 Impact resistance. An approximately 3 by 6-inch 0.010-inch thick steel test panel shall be tested according to ASTM D 2794. The coated side of the panel shall be face down over the hollow area and an impact of 80 inch-pounds shall be delivered to the side of the panel opposite the coating. The impacted area shall be evaluated for compliance with the requirements specified in 3.5.

4.3.2.4 Flexibility. Two 0.010-inch thick steel and two 0.010-inch thick steel tinplated coated panels prepared as in 4.3.2.1 shall be bent slowly (1 to 2 seconds) at room temperature (65 to 80 °F) through 180 degrees over 1/4-inch diameter mandrel with the coated side outward. Similar panels shall be conditioned for 4 hours at low temperature (0 to -5 °F). They shall be bent at that temperature over the 1/2-inch diameter mandrel through 180 degrees. The panels shall be evaluated for compliance with the requirements specified in 3.6.

4.3.2.5 Resistance to boiling water. A coated 0.032-inch thick steel panel prepared as in 4.3.2.1 shall be completely immersed in boiling distilled water for 2 hours. After cooling to room temperature, the panel shall be immediately compared with a non-immersed panel for compliance with the requirements specified in 3.7.

4.3.2.6 Inertness. Three weighed 1 by 3-inch 0.052-inch thick steel panels shall be cut from a panel prepared as in 4.3.2.1, shall be positioned vertically in a flask fitted with a condenser. The panels shall be arranged so as not to be in flat contact with each other. A 70/30 by volume mixture of VM&P naphtha and toluene shall be poured into the flask to cover the panels. The flask shall be held at 170 ±5 °F for 4 hours. The panels shall be removed, dried, cooled and reweighed. Weight shall be determined to the nearest tenth of a milligram. The gain or loss in weight of the three panels combined shall be determined and the results shall be evaluated for compliance with the requirements in 3.8.

4.3.2.7 Reflectance. Using panel prepared as in 4.3.2.1, the directional reflectance shall be determined in accordance with ASTM E 1347 and the results shall be evaluated for compliance with the requirements in 3.9.

4.3.2.8 Opacity. Apply the coating to a 2 mil wet film thickness on a black and white hiding chart.[4/] Allow to air dry for 24 hours and determine opacity in accordance with the requirements in 3.10.

4.3.2.9 Fuel and water resistance. Three 0.032-inch thick steel panels prepared according to 4.3.2.1, shall be immersed in wide mouth glass jars, liter size, each containing 350 ml of JP-8

4/ Leneta company, Ho-Ho-Kus, NJ, Form 2A, or equivalent.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:05Z
Check the source to verify that this is the current version before use.

fuel conforming to MIL-T-83133 (Jet A-1 may be used in place of JP-8), and 350 ml of distilled water. Seal the jars and place in a water bath maintained at 51 to 53 °C (124 to 127 °F) for 7 days. At the end of the test period, remove and evaluate panels for compliance with the requirements specified in 3.12.

4.3.3 Qualitative tests.

4.3.3.1 Chromium (hexavalent) content.

a. Reagents: 25 percent aqueous KOH.
b. Procedure:

(1) Add 5 ml of 25 percent aqueous KOH to 0.5 g of the extracted pigment contained in a 15 ml centrifuge tube.
(2) Agitate by shaking the tube for a few minutes, then centrifuge.
(3) The supernatant liquid should be colorless. A yellow color indicates the presence of chromate. Nonconformance to the requirement in 3.1.2 shall constitute failure of this test.

4.3.3.2 Lead content. Determine the percent of lead in accordance with ASTM D 3335 or by x-ray spectrometric analysis in accordance with the manufacturer's manual. Evaluate for compliance with 3.1.2.

4.3.3.3 Condition in container. Package condition of the coating and pigment settling shall be determined in accordance with Method 3011 of FED-STD-141. After 5 minutes of hand stirring, the coating shall be reexamined for gel bodies or undispersed pigment. The results shall be evaluated for compliance with 3.2.

4.3.3.4 Accelerated storage stability. A sealed 1 quart can of coating shall be stored for 7 days at 51 to 53 °C (124 to 127 °F). The coating shall be examined using ASTM D 1849 and evaluated for compliance to 3.3.

4.3.3.5 Volatile organic compounds (VOC) determination. Determine the VOC content in accordance with ASTM D 3960. Evaluate for compliance with the requirements in 3.11.

5. PACKAGING

5.1 Packaging and packing. The coating shall be packaged and packed in accordance with PPP-P-1892. The level of packaging and packing shall be level A, B, or C as specified (see 6.2).

5.2 Marking. Marking of each container, intermediate containers, shipping containers, and palletized loads shall be in accordance with PPP-P-1892. Special markings shall be as specified in the contract or purchase order (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:05Z
Check the source to verify that this is the current version before use.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers an organic protective coating with a maximum of 420 grams/liter (3.5 lbs/gallon) VOC to be used on the interior of steel fuel containers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (See 2.1 and 2.2).
c. Level of packaging and packing required (see 5.1 and 5.2).
d. Quantity required (see 5.1).
e. Any special marking required (see 5.2).

6.3 Subject term (key word) listing.

VOC compliant
Fuel resistant

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to previous issue due to the extensiveness of the changes.

6.5 Material Safety Data Sheets. Contracting officers should identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addressed for submission of data are listed in appendix B of FED-STD-313.

6.6 Safety/Health. Ensure that all applicable and health requirements per Federal, DoD, and Army regulations are followed during manufacturing, shipping, and handling.

Custodians:
Army - AV

Preparing activity:
Army - AV

Agent activity:
Army - ME

Review activities:
Army - MD

Project 8010-A786

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:05Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-52032D(AV)	2. DOCUMENT DATE (YYMMDD) 950406

3. DOCUMENT TITLE
COATING, INTERIOR, FUEL RESISTANT, VOC COMPLIANT FOR GASOLINE CONTAINERS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) DSN
c. ADDRESS (Include Zip Code) US ARMY AVIATION & TROOP CMD ATTN AMSAT-R-ECS 4300 GOODFELLOW BOULEVARD ST LOUIS MO 63120-1798	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:05Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:05Z
Check the source to verify that this is the current version before use.